Smart and Sustainable Built Environment

Diffusion of adoption of solar energy - a structural model analysis

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Received 19 November 2016
Revised 26 September 2017
Accepted 28 September 2017

Abstract

Purpose – The purpose of this paper is to locate determinants of the diffusion process of solar energy and examine the relationship between variables affecting diffusion and adoption solar of energy.
Design/methodology/approach – The data used in this research have been collected from users and non-users of solar energy products in Punjab. A research model was used to test for factors of diffusion and intention to use solar energy products.
Findings – The survey results show that the majority of solar product users are aware of the difference between renewable and non-renewable sources of energy. Roger's diffusion model containing five components, relative advantage, trialability, observability, complexity and compatibility, shows that relative advantage and trialability significantly influence user's intention to use solar products. In order to increase adoption and usage of solar energy products, more consideration is required toward raising the reliability and utility of solar products. Demos and trials highly convince users toward solar products. Hence sellers of solar products should give effective trials to increase acceptance of solar products.
Originality/value – This work provides a clear focus which will allow the diffusion and adoption of solar energy product by providing guidance and aspiration as well as providing platform to solar energy manufacture and government for policy making and future decision.

Keywords Adoption, Solar energy, Diffusion, Relative advantage, Renewable, Trialability

Paper type Research paper

Solar energy

Solar energy is a type of energy tackled from the energy and warmth of the Sun's beams. Researchers are working diligently to locate a reasonable, productive answer for bridling Sun-powered energy (Honsberg and Bowden, 2014). Photovoltaic cells produce power by using a silicon that change approaching daylight into power instead of warmth (photovoltaic means power from the light, photo means light and voltaic means power). Sun-powered photovoltaic cells comprise a positive and a negative film of silicon put under a dainty cut of glass. As the photons of daylight beat downward on these cells, they thump the electrons off the silicon. The contrarily charged free electrons are specially pulled into the other side of silicon cell, which makes an electric voltage that can be gathered and directed. This current is assembled by wiring the individual sunlight-based boards together in arranging to frame a Sun. The power delivered at this stage is direct current (DC) and should be changed over to alternating current (AC) reasonable for use in our home or business. The inverter turns the DC power produced by the Sun-oriented board into120-volt

Smart and Sustainable Built Environment
Vol. 6 No. 2, 2017
pp. 66-83
© Emerald Publishing Limited
2046-6099
DOI 10.1108/SASBE-11-2016-0033

AC that can be put to prompt the use by interfacing the inverter specifically to a devoted electrical switch in electrical board (Solar California, 2007).

According to The Renewable Energy Hub, sunlight-based innovation comes in numerous shapes and size and is utilized as a part of a wide assortment of business applications. Photograph voltaic types are monocrystalline silicon PV boards, polycrystalline silicon PV boards and thick-film silicon PV boards (Photovoltaics, n.d.).

World energy utilization is the aggregate energy utilized by all of human progress. Commonly measured every year, it includes all energy saddled from each energy source connected toward humankind's attempts over each and every modern and innovative division, over each nation. Being the energy source metric of progress, Wikipedia (2016) has profound ramifications for humankind's social-monetary political circle.

Organizations, for example, the International Energy Agency (IEA), the US Energy Information Administration and the European Environment Agency record and distribute energy information occasionally. Enhanced information and comprehension of world energy consumption may uncover systemic patterns and examples, which could outline current energy issues and energize development toward altogether valuable arrangements.

The IEA gauges that, in 2013, all over world energy utilization was 9,301 Mtoe, or $3.89 \times 1{,}020$ joules, equivalent to a normal energy utilization of 12.3 terawatts from 2000 to 2012; coal was the wellspring of energy with the biggest development. The utilization of oil and regular gas likewise had significant development, trailed by hydropower and renewable energy. Renewable energy developed at a rate speedier than whatever other time in history amid this period, which can be clarified by an expansion in global interest in renewable energy.

Current status of solar energy in India

Government-funded solar electricity in India was approximately 6.40 megawatt (MW) per year as of 2005. India is ranked number 1 in terms of solar electricity production per watt installed, with an insulation of 1,700-1,900 kilowatt peak (kwh/kwp); 25.10 MW was added in 2010 and 468.30 MW in 2011. According to Punjab Department of Science, Technology and Environment (2015), the cumulative installed grid connected solar power capacity is 7,568 MW and proposed target is 100,000 MW by 2022.

Solar products subsidy schemes of central government and Punjab government are Jawaharlal Nehru National Solar Mission (JNNSM), solar water heating system, subsidy scheme for solar lighting and small capacity PV systems, capital subsidy scheme for installation of 68,000 solar photovoltaic (SPV) lighting system and small capacity PV systems under JNNSM. The scheme of Ministry of New and Renewable Energy for installation of 10,000 SPV water pumping systems for irrigation purpose is implemented through NABARD. The scheme of Ministry of New and Renewable Energy for promoting SPV water pumping systems is for irrigation purpose (Abhay, 2011). Following can be observed as the benefits of solar energy over other forms of renewable energy – once introduced and paid for beginning cost venture, no extra expenses are needed; the electrical energy devoured is free.

Solar energy global status

According to Renewables Global Status Report (2015), renewable energy continued to grow in 2014, particularly in developing countries and a dramatic decline was experienced in oil price. Solar power produces a consistent supply of electricity and contributes to a reduction in global warming. The Paris Global Report (2016) reported that the annual market in 2015 was nearly ten times the world's cumulative, solar PV capacity; China, Japan and the USA gain accounted for majority of the capacity. China achieved 100 percent electrification in part because of significant off-grid solar PV installed in 2012. According to International Energy Agency (2013) despite a slowdown in 2013, Germany is expected to remain the top solar market in Europe for coming years and still boasts a quarter of the world's installed PV capacity of 26 percent, compared to 13 percent held by each of the next two countries on the list, Italy and China.

Recent trends in solar energy in India

The Indian solar market is at an interesting point at the moment. It has come from 0 to 1 GW within a period of 1.5 years. New policies have been announced and there is a palpable sense of excitement in the air. It is expected that the next year to have project allocations in excess of 2 GW. One could complain that it is difficult for an investor to keep track of developments and risks, but that drawback is much smaller than the larger benefit of finding out what works best in India. At present this lively, innovative policy space does justice to the complexity of solving the Indian power (not green power) riddle. There has been a presentation of energizing new arrangements. A movement can be seen from open to private energy buy understandings. The Sun-oriented renewable buy commitment (RPO)/renewable energy declaration (REC) business sector is failing to meet expectations (Engelmeier, 2015). According to Meisen (2006), India at this time experiencing well-built economic growth, despite the fact that million are still living in poverty without power services. Growing electrical ability is necessary. Solar energy remains a little fraction of installing ability.

Literature review

The diffusion of innovations has been a subject of viable and scholastic enthusiasm since the 1960s when the spearheading works of Fourt and Woodlock (1960), Mansfield (1961), Floyd (1962), Rogers (1962), Chow (1967) and Bass (1969) appeared. During the last 35 years, there have been several reviews of diffusion models. These include Meade (1984), Mahajan and Peterson (1985), Mahajan *et al.* (1990, 1993, 2000a, b), Baptista (2000), and Meade and Islam (2001).

It has been seen that the cumulative adoption of an innovation taken between introduction and saturation is generally modeled by an S curve. The literature has examined the reasons for cumulative diffusion S-shaped curve by developing two extreme hypotheses. One group of researchers explains that this shape is those based on the dynamics of a (broadly homogeneous) population and others are those based on the heterogeneity of the population Meade and Islam (2001). One of the first to use a heterogeneous population argument was Rogers (1962). He suggests that populations are heterogeneous in their propensity to innovate. Taking the dynamics of the population first, Bass (1969) (see A1.01) suggests that individuals are influenced by a desire to innovate and by a need to imitate others in the population.

Studies on diffusion of technology have attributed the factors affecting diffusion to various reasons. The heterogeneity of income distribution has been cited by Duesenberry (1949), Bonus, 1973) and Horsky (1990). Liebermann and Paroush (1982) provide an economic argument that income heterogeneity, price and advertising are important drivers of the diffusion process. The time at which an innovative product is introduced may have an effect on the rate of its diffusion (Kohli *et al.*, 1999). Van den Bulte (2000) found that increased purchasing power, demographic changes and the types of product determine the change in the speed of diffusion of innovations. Wareham *et al.* (2004) examined socio-economic factors underlying the diffusion of the internet and 2G mobiles in the USA. In an empirical study, Baptista (2000) found that regional effects have significant influence on the rate of diffusion.

Tanner (1974) incorporated environmental variables (GDP/capita) within the diffusion model. Lee (1990) in his empirical cross-sectional study of 70 nations investigated the innovativeness of nations. The significant predictor variables were GNP/capita, proportion literate, proportion of scientists and the proportion of GNP generated by the manufacturing sector. Kiiski and Pohjola (2002) identified that the main determinants of diffusion are GDP per capita and access cost.

Some authors have made the saturation level (market potential) a function of price, advertising or some other measure of market activity (Mahajan and Peterson, 1978). The impact of price on market potential has been studied by Mahajan and Peterson (1978), Bass (1980), Bass and Bultez (1982), Kalish (1985) and Horsky (1990). Jun and Park (1999), Jun *et al.* (2002) and Horsky and Simon (1983) examined the effect of advertising on the probability of adoption. Robinson and Lakhani (1975) were the first to include price impact in the Bass model. Kamakura and Balasubramanian (1988) found that price affected adoption probability for relatively highly priced goods. Gruber and Verbovan (2001) found that competition between suppliers increased the diffusion rate.

Kalish and Lilien (1983) examined the relevance of price subsidy for accelerating the diffusion of beneficial technologies (such as alternative energy systems). In most cases, they find that the subsidy should decrease as diffusion accelerates. Desiraju *et al.* (2004) found that per capita expenditure was positively related to the rate of adoption, while higher prices decreased adoption rates.

India occupies fifth place in the world these days in installing solar energy. Spread of renewable energy output has turned over less than half as of 2014 (Nagamani *et al.*, 2015). Coal accounts for close to maximum of India's whole energy use, followed by oil and very less as natural gas (Yep, 2011). To overcome the energy and peaking shortage, alternative energy sources are needed to provide reliable quality power at reasonable rates with consumer interests (National Electricity Policy, 2005). India is the world's sixth largest energy consumer. The majority of people can access electricity. Yet there is power deficiency and blackout mostly in rural residential districts. The electricity generated is below the demand (Pranti *et al.*, 2013). India has swiftly developed its solar power sector. Cheaper solar power would help thousands of Indian homes gain access to electricity for the first time (Dutta, 2012). The extensive application of solar energy in rural area has positive social and health effect (Limao *et al.*, 2012).

The initial price of solar energy and electricity cost are two most significant limits that make the use of alternative of energy. Friends and family members are the principal origin of information and awareness regarding to fix SWHS (Ali *et al.*, 2009; Kumaresh and Praveena, 2012). Fund allocation and actual regional suitability for solar energy are identified as potential mismatch of solar energy (Razykov *et al.*, 2011). The company creates positioning in the mind of people that SWH is cost effective, environment friendly and maintenance cost is less. Both users and non-users felt that warranty period was less for SWH. The bulk of the non-user was willing to buy SWH, if the initial cost of the product is scaled down (Jacobsson and Johnson, 2000). NABARD and Reserve Bank of India develop model scheme and associated facilitative normative, strategy and restrictions for suitable capital cost to promote contribution in rural power supply (National Rural Electrification Policies, 2006). Governmental and institutional finance put important influence on the decision to adopt PV power supply system in developing countries (Meisen, 2006).

The unexpected decrease in residential solar usage because of polices can encourage solar provider to set up business within the country and at local level (Kerschen, 2007). Within the country, it is better to recognize and associate with national discrepancies in solar energy's acceptance (Zahran *et al.*, 2008). Their effort has been made to summarize the availability, current position, strategies, perspective promotional polices, major achievement and future potential solar energy option in India (Sharma *et al.*, 2012). Consumers are just satisfied with the solar home service system and little gap exists between exception and perception of the consumer (Momotaz and Karim, 2012). SPV technology can be implemented in India as solar resources to fill up long-term global energy crises (Basu *et al.*, 2015). There is a great opportunity for India to be a manufacturing hub. The greatest challenge for government is to have channelized sufficient investment in the sectors (Sharda, 2015).

The energy-conserving strategies were diffused rapidly throughout the housing technological and economic condition (Hirshberg and Schoen, 1974). The weak institutional policy support, growth of electricity use and highly fossil fuel set back the diffusion (Pfeiffer and Mulder, 2013). The financial sector has not made any effect in the dissemination process. Most of the consumers use payback period and not the net present value of decision making (Rai and Sigrin, 2013). Diffusion spreads new ideas and fresh thoughts and results in altering the traditional social system through innovation, communication and so on (Rogers, 1983). The government should launch subsidies and speed up the diffusion of SPV in rural areas which can be helpful in agriculture sector (Bankanga, 2011). Paul and Uhomoibhi (2012) recommended that for significant penetration of solar electricity in rural areas and urban areas without reliable grid electricity training should be provided to skillfully qualified solar electrical technicians, in both urban and rural areas. Renewable (solar) energy courses should be introduced to all people, at all education levels and through all possible modes of education. The government should put policies and programs in place that will maximize private, community and public investment in rural solar electricity. The government should support policies that encourage community investment in stand-alone solar systems including community finance funds and training. Ndzibah (2010) pointed out that solar solution as an alternative source of energy was hardly identified as viable option due to lack of adequate knowledge and information. They observed that the public seem to hardly understand the real components that make up a complete solar solution.

Diffusion model

Technology acceptance model (TAM) and innovation diffusion theory (IDT) are two major adoption and intention models developed to assess the determinants of user's intention to use technology (Ajzen and Fishbein, 1980; Davis, 1986). TAM model theorizes that perceived usefulness (PU) and perceived ease of use (PEOU) are the primary determinants of system use. Roger's IDT is a major breakthrough in this area which postulates that innovation diffusion is achieved through users' acceptance and use of new ideas or things (Zaltman and Stiff, 1973). Rogers (1995) stated that an innovation's relative advantage, compatibility, complexity, trialability and observability were found to explain 49-87 percent of the variance in the rate of its adoption. Tornatzky and Klein (1982) found that only relative advantage, compatibility and complexity were consistently related to the rate of innovation adoption. TAM and IDT are among the most influential theories in explaining and predicting system use and innovation adoption. TAM and IDT have some noticeable similarities. The relative advantage construct in IDT is often viewed as the equivalent of PU construct in TAM, and the complexity construct in IDT is very similar to the PEOU concept in TAM (Moore and Benbasat, 1996; Davis *et al.*, 1989; Karahanna *et al.*, 1999). IDT model was applied for present study as it is more comprehensive and incorporates all factors of TAM model.

Diffusion predicts five behaviors that decide the success of an invention. At the relative advantage stage, user's belief of an innovation is better than the idea it supersedes. At compatibility stage, consumer is consistent with reviewing value, past experience and need of potential. At complexity stage, buyer is supposed as disparity to be familiar with and make use of. At trialability stage, they may be well-informed with a limited basis. At the last stage, results of the innovation are observable to some others (Harder, 2011).

Peterson's (2010) analysis shows that adapter comes after five phases of decision making. In the beginning phase of knowledge, an individual is endangered to the innovation accessible to him and how much it is purposeful. In the second phase, persuasion comes when an individual moves from the positive and adverse approach toward the founding.

In the third stage, decision arises when an individual takes an action that leads to alternative, i.e. either live with or pass up the founding. In the fourth phase of implementation, an individual puts the innovative through utilizing it. In the fifth stage, confirmation occurs when an individual's facial expression for handling an innovative option is finished.

Leek (1997) studied the diffusion adaptation theory that identifies five categories of adopters based on consumer personality and behavior. In the first category, innovators are less adaptors; in a second category, they are early adopters more than innovators; in the third category early majority; in the fourth category, Late majority were maximized. In fifth category, laggards were one third. Half of the consumers being first three stages.

Considering the significant potential contribution of solar energy in overcoming the energy shortage in India, the present study seeks to identify significant factors that affect diffusion and adoption of solar energy in India.

Objectives

(1) To identify the determinants of diffusion of solar energy in Punjab.

(2) To examine the relationship between variables affecting diffusion and adoption of energy and user's intention to continue to use solar energy-powered products.

Methodology

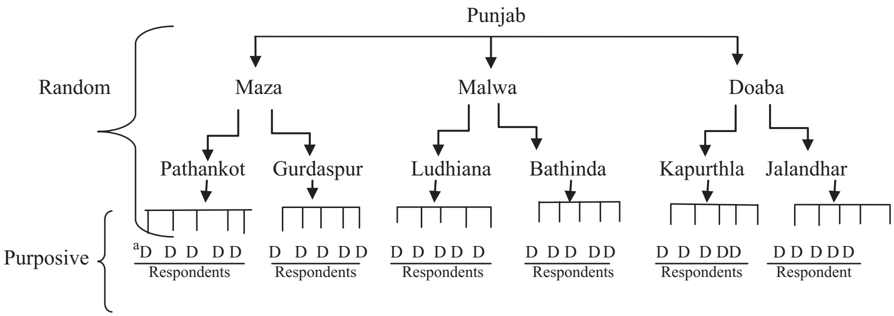
The survey strategy was adopted for the present study. The choice of a survey strategy for present research is guided by research objectives, available literature, the amount of time and other resources available. The population of the present study includes users of solar energy-powered products in Punjab. Punjab is a power deficit state. Punjab's demand for power was 48,881 MU and supply was 39,918 MU, falling short of 18.3 percent in year 2011-2012 during peak days (Gupta and Singh, 2012). Highest SPV pumps were installed in Punjab in year 2011-2012 (Ministry of Statistic and Programmed Implementation, Government of India, 2012). However, the installation of SPV system during 2011-2012 was zero in Punjab (Annual Report, MNRE, 2011-2012). Punjab is the third last state in terms of solar energy capacity in India (Jawaharlal Nehru Solar Mission, 2012-2013) and third last North Indian state with installed generating capacity of electricity utilities (Ministry of Statistic and Programmed Implementation, Government of India, 2012).

Considering the ample need for increasing installed capacity, Punjab needs to resort on solar energy for bridging the gap between demand and supply. Hence, Punjab was selected for the present study.

Punjab is divided into three zones: Maza, Malwa and Doaba. Two cities were selected from each of these zones based on cluster sampling. Pathankot and Gurdaspur from Maza; Ludhiana and Bathinda from Malwa; and Kapurthala and Jalandhar from Doaba were selected.

From each of these cities, five solar product dealers registered with Punjab Energy Development Agency were selected (see Figure 1). A total of 300 users were selected from the database maintained by selected dealers. Data were analyzed through Statistical Package for the Social Sciences and Analysis of Moment Structures (AMOS).

The survey instrument was pretested and examined for validity and reliability. To ensure content's validity of questionnaire items, discussions were held with two experts having more than ten years of experience in solar energy sector and two professors from the academic field. After two revisions, consensus was formed among the suggestions of experts and questionnaires were administered to 60 residents (based on convenience sampling) from Pathankot districts for assessing reliability. Reliability for scale used in the survey was assessed through Cronbach's α.



Note: [a]Dealers

Figure 1.
Cluster sampling of study area

Reliability of the survey instrument was ascertained through Cronbach's α reliability coefficients. All constructs of IDT showed good reliability, as Cronbach's α values were more than 0.70, which indicates acceptable level of reliability (Cronbach, 1951; Cortina, 1993; Peter, 1979).

Data analysis

This section empirically evaluates the understanding of respondents about renewable and non-renewable energy sources. The sources that influence user's adoption of solar energy-powered products were identified through rank order method. Finally, a diffusion model based on IDT of E. Rogers was applied to identify the major drivers of solar energy diffusion.

Table I shows that the majority (96.0 percent) of the solar product users are aware of the difference between renewable and non-renewable sources of energy. Therefore, there is high knowledge amongst users about the renewable and non-renewable energy sources.

Various sources that influence user's adoption of solar energy-powered products are shown in Table II. Family and friends and newspaper are major sources that influence user's adoption of solar energy-powered products followed by television, books, retailers and radio. Sources that have least influences are environmental organizations and manufacturers followed by existing users, government agencies and internet.

A research model for diffusion and intention to use solar energy-powered products

In order to examine the drivers of user's intention to use solar energy products, Everett Rogers diffusion scale was used. Rogers argues that diffusion is the process by which an innovation is communicated over time, among the participants in a social system. Confirmatory factor analysis (CFA) was used to test the measurement model. Measurement model consists of six latent constructs, i.e. relative advantage, compatibility, complexity, observability, trialability and intention to use. All these latent constructs were measured by indicator variables/statements shown in Table III. AMOS software was used to conduct CFA. For assessing an overall model fit, fit statistics assessed include: χ^2, df, p-value;

Table I.
Knowledge of the difference between renewable and non-renewable energy sources

S. No.	Response	Frequency	Percent
1	Yes	288	96
2	No	12	4

Source: Compiled from SPSS output

Sources	Ranks 1	2	3	4	5	6	7	8	9	10	11	12	Score	Ranks
Friends and family	63	69	41	25	13	11	16	9	10	6	4	33	1,329	1
Television	25	34	23	75	39	27	10	17	21	3	24	2	1,532	3
Newspaper	37	13	65	46	35	17	16	10	17	33	7	4	1,519	2
Radio	17	19	23	36	19	52	26	18	24	53	3	10	1,900	6
Books	14	52	16	18	32	30	57	38	12	9	20	2	1,723	4
Retailers	28	33	22	5	8	32	59	53	20	15	10	15	1,869	5
Govt. agencies	9	6	13	28	31	64	8	53	52	11	13	12	2,056	8
Env. org.	0	2	46	16	27	14	16	39	46	53	12	29	2,273	11
Manufacturers	2	25	9	36	37	7	35	4	32	39	55	19	2,238	10
Internet	63	29	15	7	24	12	9	11	17	34	55	24	1,923	7
Existing users	37	23	18	27	13	11	6	14	36	10	72	33	2142	9
Others	0	3	15	17	18	23	21	40	27	18	55	63	2,598	12

Source: Compiled from SPSS output

Table II. Sources influencing user adoption of solar energy-powered products

Item code	Scales/items	Constructs
R1	Solar energy-based products are reliable and efficient	Relative advantage
R2	Solar energy-based product helps me to better manage my energy requirements	
CP1	After using solar energy-based product, I need not to try other forms of energy	Compatibility
CP2	I worry about the necessary information required for using solar energy-based product	
CP3	It obligates me to use solar energy-based product when I could not do my work through other forms of energy (e.g. electric cuts)	
CP4	The solar energy-based product has less voltage fluctuations as compared to conventional source of energy	
CX1	Solar energy-based product are difficult to understand for non-technical person	Complexity
CX2	It is easy to use solar energy-based product even if one has not used them before	
CX3	Solar energy-based products are convenient to understand	
OB1	I observed others using solar energy-based product and saw the advantages of doing so	Observability
OB2	The information about solar energy-based product is easy to find	
OB3	I was influenced by what I observed as the benefits of using solar energy-based product	
TR1	A demo of solar convinced me that using solar technologies was better than using other sources of energy	Trialability
TR2	I do not need a trial to be convinced which solar energy-based product are the best for me	
TR3	It was easy to use solar energy-based product more frequently after trying them out	
TR4	It did not take me much time to try solar-based product before I finally accepted their use	
ITU1	Solar technology usage is appropriate for my working style and I will continue using it	Intention to use
ITU2	What I have observed about the use of solar technology in my region will make me keep using them	
ITU3	The benefits of solar technology will make me continue to use them in the future	
ITU4	Trying out the opportunities of using solar technology will make me continue using them in future	
ITU5	I intend to continue to use solar technology because it manages my needs better	
ITU6	Because solar technology are appropriate to my home I will use them in future	
ITU7	The ease of use of solar technology will make me continue to use them	

Source: Compiled from survey instrument

Table III. Factors of diffusion and intention to use solar energy-powered products absolute fit measures: root mean square error of approximation (RMSEA), standardized root mean residual (SRMR) normed χ^2; normed fit index, incremental fit indices: comparative fit index (CFI) and parsimony fit index: adjusted goodness-of-fit index (AGFI).

Reliability and validity of model

Reliability and convergent validity of constructs used in the measurement model is shown in Table IV. Reliability of constructs was estimated through Cronbach's α. The value of α coefficient varies from 0 to 1, and a value of 0.6 or less generally indicates unsatisfactory internal consistency reliability (Malhotra, 2008). The generally agreed upon lower limit for Cronbach's α is 0.70 (Hair *et al.*, 2013). Reliability value for all factors varies from 0.806 to 0.858. Therefore, all the constructs show acceptable level of reliability scores.

For validation of measurement model, convergent and discriminant validity scores were evaluated. Convergent validity of the measurement model was assessed through factor loadings, average variances extracted (AVE) and construct reliability. Construct reliability for all the factors in the measurement model was estimated above 0.70 which shows that the measures all consistently represent the same latent constructs, i.e. internal consistency exists. All items load highly on their respective factors and have loadings above 0.70 confirming the convergent validity of the constructs (Hair *et al.*, 2013). Item 1, item 2 and item 4 of construct intention to use were dropped as the values of their factor loadings were below the recommended level. Convergent validity is found to be adequate when constructs have AVE of at least 0.5 (Fornell and Bookstein, 1982; Gefen *et al.*, 2000; Wixom and Watson, 2001). The AVE were estimated for constructs and found to be above the recommended level.

Table V shows shared variance and AVE estimating the discriminant validity of the factors. Discriminant validity is supported when the AVE for a construct is greater than the shared variance between constructs (Hair *et al.*, 2013). For satisfactory discriminant validity, the AVE for each construct should be greater than the variance shared between the

Construct code	Constructs	Item code	Factor loading	Cronbach's α reliability	Average variance extracted	Construct reliability
RA	Relative advantage	R1	0.81	0.806	0.52	0.80
		R2	0.83			
CP	Compatibility	CP1	0.85	0.842	0.77	0.85
		CP2	0.76			
		CP3	0.70			
		CP4	0.73			
CX	Complexity	CX1	0.77	0.858	0.68	0.86
		CX2	0.92			
		CX3	0.77			
OB	Observability	OB1	0.78	0.841	0.65	0.85
		OB2	0.80			
		OB3	0.83			
TR	Trialability	TR1	0.82	0.849	0.92	0.90
		TR2	0.87			
		TR3	0.76			
		TR4	0.86			
ITU	Intention to use	ITU3	0.72	0.841	0.80	0.86
		ITU5	0.90			
		ITU6	0.70			
		ITU7	0.76			

Source: Compiled from SPSS and AMOS output

Table IV. Reliability and validity of measurement model construct and other constructs in the model (Fornell and Bookstein, 1982; Gefen *et al.*, 2003; Wixom and Watson, 2001). The analysis reflects that the shared variance between factors were lower than the AVE of the individual factors, confirming the adequate discriminant validity (see Table V, Figure 2).

Table VI shows model fit statistics for the structural model for diffusion and intention to use solar energy-powered products from structural model output. The overall model χ^2 is 231.539 with df = 155. The χ^2 is significant ($p < 0.001$), which means that the observed covariance matrix does not match estimated covariance matrix. This is due to the large sample size ($n = 300$); however, the normed χ^2 is within the suggested guidelines at 1.494 (Hair *et al.*, 2013). The value for RMSEA, an absolute fit, is 0.041, which is below the 0.08 guideline for a model with 20 observed variables and a sample size of 300 (Hair *et al.*, 2013). The value of SRMR is 0.039, which is below the conservative cutoff value of 0.05 (Hair *et al.*, 2013). Incremental fit indices reflect a good model fit, as CFI value is 0.973,

Factor	RA	CP	CX	OB	TR	ITU
Relative advantage (RA)	0.52					
Compatibility (CP)	0.17	0.77				
Complexity (CX)	0.01	0.01	0.68			
Observability (OB)	0.02	0.04	0.02	0.65		
Trialability (TR)	0.03	0.02	0.04	0.155	0.92	
Intention to use (ITU)	0.05	0.05	0.04	0.06	0.35	0.80

Notes: Diagonal elements are the average variance extracted; off-diagonal elements are shared variance
Source: Compiled from AMOS output

Table V. Shared variance and average variance extracted

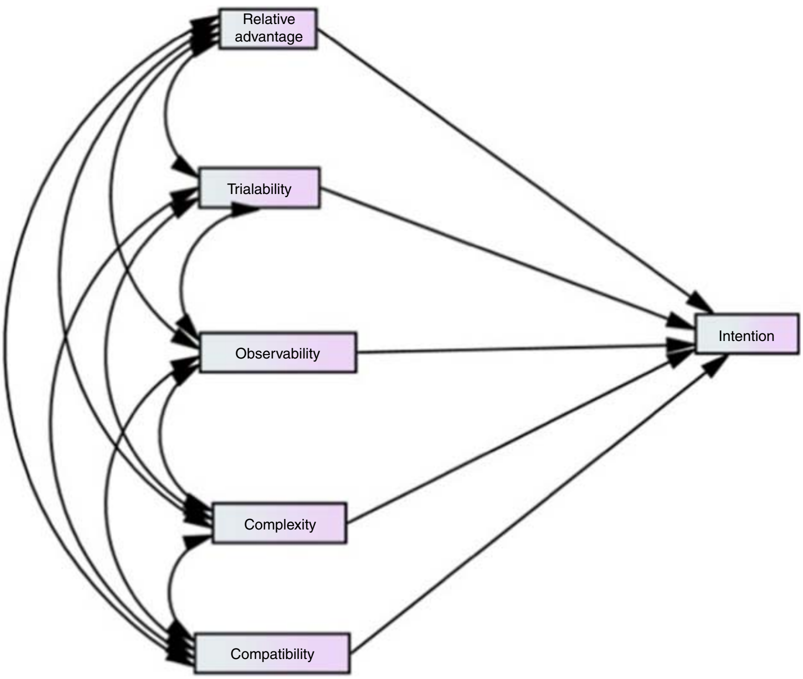


Figure 2. Diffusion model of solar energy

which is above suggested cutoff value of 0.90 (Hair *et al.*, 2013). The parsimony index of AGFI has a value 0.905, which reflects good model fit (Hair *et al.*, 2013).

The results of hypothesis tested are shown in Table VII. From the table, it can be concluded that amongst five exogenous constructs only two constructs (relative advantage and trialability) have significant impact on the user's intention to use solar energy-powered products.

The first null hypothesis that there is no significant impact of relative advantage on the user's intention to use solar energy-powered products was rejected, $t = 1.982$, $\beta = 0.125$, $p = 0.047$. Therefore, the diffusion variable of relative advantage has significant influence on the user's intention to use solar energy-powered products. The second null hypothesis that there is no significant impact of compatibility on the user's intention to use solar energy-powered products was accepted, $t = 0.518$, $\beta = 0.031$, $p = 0.605$. Therefore, the diffusion variable of compatibility has no significant influence on the user's intention to use solar energy-powered products. The third null hypothesis that there is no significant impact of complexity on the user's intention to use solar energy-powered products was accepted, $t = 0.219$, $\beta = 0.013$, $p = 0.826$. Therefore, the diffusion variable of complexity has no significant influence on the user's intention to use solar energy-powered products. The fourth null hypothesis that there is no significant impact of observability on the user's intention to use solar energy-powered products was accepted, $t = 0.482$, $\beta = 0.031$, $p = 0.630$. Therefore, the diffusion variable of observability has no significant influence on the user's intention to use solar energy-powered products. The fifth null hypothesis that there is no significant impact of trialability on the user's intention to use solar energy-powered products was rejected, $t = 7.177$, $\beta = 0.508$, $p = 0.001$. Therefore, the diffusion variable of trialability has significant influence on the user's intention to use solar energy-powered products.

Table VI.
Fit indices for structural model

χ^2	Absolute fit measures	Incremental fit indices	Parsimonyfit indices
χ^2: 231.539 $p = 0.000$ df: 155	RMSEA: 0.041 SRMR: 0.039 Normed χ^2: 1.494	CFI: 0.973	AGFI: 0.905

Source: Compiled from AMOS output

Table VII.
Summary of hypothesis testing

Hypothesis	Standardized regression weight (β)	t-value	p-value	Result
There is no significant impact of relative advantage on the user's intention to use solar energy-powered products	0.125*	1.982	0.047	Rejected
There is no significant impact of compatibility on the user's intention to use solar energy-powered products	0.031	0.518	0.605	Accepted
There is no significant impact of complexity on the user's intention to use solar energy-powered products	0.013	0.219	0.826	Accepted
There is no significant impact of observability on the user's intention to use solar energy-powered products	0.031	0.482	0.630	Accepted
There is no significant impact of trialability on the user's intention to use solar energy-powered products	0.508**	7.177	0.000	Rejected

Notes: *,**Significant at 0.05 and 0.001 levels, respectively
Source: Compiled from AMOS output

Chang and Tung (2008) and Lee *et al.* (2011) combined TAM and IDR to study student's behavioral intention to use the online learning course websites and found that compatibility, PU, PEOU, perceived system quality and computer self-efficacy were critical factors for students' behavioral intentions to use the online learning course websites. Results of the present study are partially consistent with the findings of Tornatzky and Klein (1982) who found that only relative advantage, compatibility and complexity were consistently related to the rate of innovation adoption and Ntemana and Olatokun (2012) who identified that complexity, observability and relative advantage were revealed to be significant attributes that positively affect attitude of the lecturers toward using technology, whereas only relative advantage and trialability have shown major significance on intention to use in the present study. One possible explanation of this relationship may be that present findings are specific to solar energy, whereas results of Tornatzky and Klein (1982) and Ntemana and Olatokun (2012) study were generalized on overall innovation adoption and non-energy-specific area, respectively.

Conclusion

Majority of users know the difference between renewable and non-renewable energy sources. Family, friends and newspapers are major sources that influence user adoption of solar energy-powered products, followed by television, books, retailers and radio. Roger's diffusion model, containing five constructs of relative advantage, trialability, observability, complexity and compatibility, shows that only relative advantage and trialability significantly influence user's interaction to use solar energy-powered products. Therefore, in order to increase adoption of usage of solar energy-powered products, more consideration is required toward raising the reliability and utility of solar energy-powered products. Demos and trials highly convince users toward usage of solar energy-powered products. Hence, seller of solar energy-powered products should give effective trials and demos to increase acceptance of solar energy-powered products.

Cheng and Townsend (2000) and cited four issues of compatibility with using information and communication technology (ICT). Pelgrum (2001) identified lack of knowledge and skills about ICTs as the main barrier of diffusion. Law and Plomp (2003) hints at Rogers's (2003) characteristic of trialability as the major factor in ICT diffusion. Fitzgerald *et al.* (2002) found lack of professional development as one of the key failures for diffusion of ICT. Rodrigo (2005) found that the inability of the end users to practice (trialability) using the innovation as the major reason for hindrance in diffusion of innovation. Pelgrum (2001) indicates that not seeing others use (observability) the ICT innovation may be an obstacle to continued use. Richardson (2009b) found complexity as a dominant factor in respondent's choice to adopt ICT innovation in Cambodia. Solar power epitomizes an enormous resource which could address the world's needs for clean power generation. Further research is needed to confirm the drivers of diffusion of solar energy extracted by the present study.

Scope for future research

The present study provides a reliable and valid IDT model in the context of solar energy diffusion and adoption in India. Review of literature reveled that vast amount of literature is available on the use of TAM, IDT and other models; however, there is dearth of studies utilizing technology acceptance and diffusion models in the area of solar energy. Similar, studies in emerging markets can reveal additional factors that are significant in diffusion and adoption of solar energy. For generalization of results of the present study outside Punjab, study may be replicated at national and cross-country levels.

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